

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS 2026 0 QUESTIONS 0% WEIGHTAGE

↓100%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Trigonometric Equations - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Trigonometric Equations - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in Kcet Trigonometric Equations.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The trigonometric function $y = \tan x$ in the II quadrant..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If $\cos x = |\sin x|$ then, the general solution is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is

- (A) -1
- (B) 1
- (C) 0
- (D) 2

Correct Answer: None

Solution Analysis: $\cos x + \cos^2 x = 1 \Rightarrow 1 - \cos^2 x = \cos x \Rightarrow \sin^2 x = \cos x \Rightarrow \sin^2 x + \sin^4 x \Rightarrow \cos x + (\cos x)^2 = 1$